# Daniel Aarao Reis Arturi

**Research Scientist and Research Engineer**

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## RESEARCH PROFILE

Research scientist and engineer building reproducible benchmarks, agentic systems, and experimental infrastructure for open-ended problems. Current work spans AI for mathematics, mechanistic interpretability, and agentic query optimization for vector search.

**Research interests** AI for mathematics, agentic systems, database systems, query optimization, vector search, mechanistic interpretability, and reproducible evaluation

## RESEARCH EXPERIENCE

### NYU Tandon School of Engineering, VIDA Center — May 2026-Present

*Research Scientist* | DARPA expMath | Lea | Program

- Helped develop Lea, an open-source platform for mathematician-controlled formalization that supports proof decomposition, mid-run intervention, trajectory inspection, manual correction, and auditable Lean 4 output.
- Worked on the agentic harness and its research infrastructure, integrating project-aware execution, configurable sub-agent roles, Lean checking, typed event streams, persistent proof history, and shared state across standalone and Overleaf clients.
- Built the React/Vite/TypeScript and FastAPI research interface and developed the Overleaf-to-Lean workflow, including dependency-aware formalization, counterexample handling, self-repair, theorem-specific review, and GitHub export.
- First-authored the Lea paper accepted to the Sixth Workshop on Mathematical Reasoning and AI at NeurIPS 2026; contributed related-work synthesis, research framing, architecture documentation, case-study analysis, and technical writing.

### New York University, Courant Institute of Mathematical Sciences — Sep 2026-Present

*Independent Study Researcher* | Database Systems | Prof. Jinyang Li

- Building FANNBench, a cross-system research benchmark for filtered approximate nearest-neighbor search across PostgreSQL/pgvector, Elasticsearch/Lucene, and Weaviate, with deterministic workload translation, exact filtered top-*k* conformance, same-execution telemetry, and controls separating native execution, observation, and policy overhead.
- Designing the next stage: an offline language-model agent that will synthesize lightweight, workload-specific early-termination policies from prior query traces. Generated policies will make continue-or-stop decisions inside existing HNSW search paths with no model call at query time; implementation and held-out evaluation are forthcoming.

### Algoverse AI Research — Apr 2025-Present

*Research Intern* | AI Safety and Mechanistic Interpretability | Paper | Project | Code

- Studied emergent misalignment through the parameter geometry of LoRA-adapted Qwen2.5-7B and Llama 3.1-8B models. Across harmful tasks, mean principal angles were about 20 degrees versus an approximately 85-degree random baseline, with projection overlap near 0.8.
- Used linear mode connectivity to test functional equivalence across independently fine-tuned models; cross-task interpolations retained misaligned behavior while preserving more than 90% coherence. Extended the work to post-hoc and during-training interventions and GSM8K retention.
- Co-authored the NeurIPS 2025 workshop paper as a joint first author and began a separate study of how multi-thousand-turn conversations and persistent memory affect model persona stability.

### McGill University and University of Passau — Aug 2025-Mar 2026

*Research Intern* | Graph Database Systems | Artifact | Earlier preprint

- Adapted Dewey and PrePost structural indexes to tree-shaped property-graph data and evaluated baseline traversal and indexed variants across Apache AGE, Kuzu, and Neo4j using 100,000-node synthetic trees and LDBC SNB workloads.
- Measured 6.7x to more than 300,000x microbenchmark speedups and 1.1x to 16x gains on three LDBC queries in AGE, while documenting mixed results in native graph engines and index-maintenance costs; released a Dockerized reproducibility artifact and co-authored the vision paper accepted to the EDBT 2027 Research Track.

### McGill University, Distributed Information Systems Group — Jan-Jun 2026

*Research Intern* | Quantum Computing for Data Systems | Paper | Artifact

- Formulated fault-tolerant distributed data allocation through binary-slack, calibrated unbalanced-penalty, and domain-wall QUBO encodings, with exact ILP and brute-force oracles for feasibility and optimality checks.
- Evaluated D-Wave simulated annealing, hybrid CQM, Advantage hardware, and QAOA while separating QPU time from end-to-end cloud overhead. The study found no quantum advantage, mapped current scaling barriers, and produced a first-author workshop paper accepted at VLDB 2026 QC&DKM and delivered as an oral presentation.

## PUBLICATIONS AND MANUSCRIPTS

*Lea: An Agent Backbone for Mathematician-Controlled Formalization in Lean*
**Daniel Aarao Reis Arturi**, Shaswat Patel, Kevin Gomez, Jaume de Dios Pont, Jorge Poco, Eduardo V. Teixeira, Sam Westrick, Claudio Silva, and Chinmay Hegde. Accepted to the Sixth Workshop on Mathematical Reasoning and AI (MATH-AI), NeurIPS 2026. First author. [Code]

Open-source Lean formalization backbone shared by LeaChat and LeaOverleaf, with a case study formalizing the Durfee square generating function under a mathematician's guidance.

*Seeing the Trees for the Forest: Leveraging Tree-Shaped Substructures in Property Graphs*
Accepted to the EDBT 2027 Research Track. **Co-author.** [Artifact] [Earlier preprint]

Implemented and benchmarked structural indexes across Apache AGE, Kuzu, and Neo4j and built the Dockerized reproducibility artifact. Selected for the 2026 McGill Science Undergraduate Research Showcase.

*Shared Parameter Subspaces and Cross-Task Linearity in Emergently Misaligned Behavior*
**Daniel Aarao Reis Arturi***, Eric Zhang*, Andrew Ansah, Kevin Zhu, Ashwinee Panda, and Aishwarya Balwani. arXiv:2511.02022, 2025. Joint first author. [Project] [Code]

* Joint first authors with equal contributions; authors listed alphabetically. Expanded manuscript: *Of Basins and Vaccines: Emergent Misalignment Through the Lens of Parametric Geometry* (ICLR submission).

*Evaluating a Quantum Solution: How hard can it be?*
**Daniel Aarao Reis Arturi**, Emilie Fahber, Stefanie Scherzinger, and Bettina Kemme. Accepted at the 2nd Workshop on Quantum Computing and Data/Knowledge Management, VLDB 2026. First author and oral presenter. [Artifact]

Experience report and reproducibility artifact covering QUBO formulation, simulators, hybrid solvers, D-Wave Advantage hardware, QAOA, classical baselines, and practical scaling limits.

## SELECTED RESEARCH SYSTEMS AND STUDIES

### Profiling and Optimising Energy Efficiency in GFlowNet Molecular Generation

*Python, PyTorch, CUDA, CodeCarbon, Slurm* [Code] [Report]

- Instrumented a ~600M-parameter GraphTransformerGFN with timing-only, CodeCarbon, and CUDA-synchronized measurement modes at less than 0.2% overhead; found molecular sampling consumed 94.4% of step time while GPU utilization remained below 13%.
- Parallel sampling with four workers improved throughput about 1.9x and reduced total energy use by approximately 47%; automated Slurm sweeps covered batch sizes 64, 128, and 256 across three seeds and all measurement modes.

### NYC Housing Research

*Python, FastAPI, SQLite, retrieval and evaluation* [Code]

- Built a local-first cited research system over official NYC legal sources, separating keyless citation search from optional semantic indexing and evidence-cited generation while preserving source versions, provenance, explicit coverage gaps, and per-resource provider consent.
- Developed retrieval and answer-evaluation infrastructure, citation-support checks, report-bound human-review workflows, explicit provider-cost approvals, and documented gates separating software regression evidence from substantive housing-law review.

## EDUCATION

**New York University, Courant Institute of Mathematical Sciences** | M.S. Computer Science | Fall 2026-Present
**McGill University** | B.Sc. Honours Software Engineering | May 2026 | GPA 3.99/4.0 | First Class Honours
**Selected coursework** Graduate Distributed Systems, Graduate Applied ML, Graduate Computer Networks, Sustainability for ML, Algorithms, Operating Systems, Database Systems, Probability and Statistics

## AWARDS AND RECOGNITION

**NeurIPS 2026 MATH-AI** Workshop Paper Acceptance | First Author
**NeurIPS 2025 workshops** UniReps - Oral Award | MechInterp - Honourable Mention | CogInterp - Poster Acceptance
**VLDB 2026 QC&DKM** Workshop Paper Acceptance | First-Author Oral Presentation
McGill Science Undergraduate Research Showcase (2026) | Wing Hing Chan Scholarship in Science | Algoverse Merit Scholarship (twice) | McGill Dean's Honour List

## RESEARCH METHODS AND TECHNICAL SKILLS

**Methods** Experimental design, benchmark design, reproducible evaluation, filtered-ANN evaluation, query optimization, performance instrumentation, exact-oracle validation, mechanistic interpretability, parameter-subspace analysis, linear mode connectivity, QUBO, statistical analysis
**Research stack** Python, PyTorch, Transformers, NumPy, pandas, scikit-learn, Jupyter, CUDA, Slurm, CodeCarbon, D-Wave Ocean, Qiskit Aer
**Systems** Lean 4, Mathlib, FastAPI, React, TypeScript, SQLite, PostgreSQL, pgvector, Elasticsearch/Lucene, Weaviate, Docker, Git, Apache AGE, Kuzu, Neo4j